

Bangladesh University of Business and Technology (BUBT)



LAB Report

Course Code :
Course Title :
Experiment No. :
Experiment Name :

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Intake : 53
Section : 01
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Date of Submission :

Signature of Teacher

Experiment Name: VLAN Configuration

Introduction:

A Virtual Local Area Network (VLAN) allows devices to be grouped logically, regardless of their physical location. VLANs reduce broadcast traffic, improve network performance, and enhance security by isolating different groups. In this experiment, VLANs were configured on four switches using Cisco Packet Tracer. Devices were assigned to VLAN 10 (Gamer), VLAN 20 (Coder), and VLAN 30 (Seller), and inter-switch communication was established using trunk links.

Software and Tools Used:

1. Cisco Packet Tracer
2. Four Layer-2 Switches
3. PCs and Laptops
4. Ethernet Straight-through and Crossover Cables
5. Command Line Interface (CLI)

Procedure:

Manual Procedure (Switch 1 and Switch 2):

1. Open Cisco Packet Tracer and create a new project.
2. Place Switch 1 and Switch 2 and connect PCs and Laptops according to VLAN assignment.
3. Assign IP addresses to devices as per VLANs:
 - ◆ VLAN 10 (Gamer): 192.168.56.1 – 192.168.56.2
 - ◆ VLAN 20 (Coder): 192.168.67.1 – 192.168.67.2
 - ◆ VLAN 30 (Seller): 192.168.57.1 – 192.168.57.2
4. Connect Switch 1 fa0/7 ↔ Switch 2 fa0/5 using a crossover cable.
5. Access CLI on both switches, create VLANs, and assign ports to VLANs.
6. Enable trunking on inter-switch links.
7. Test connectivity between devices in the same VLAN using ping.

Command Procedure (Switch 3 and Switch 4):

1. Enter privileged EXEC mode using "enable" and global config using "configure terminal".
2. Create VLANs and assign names:
 - ◆ "vlan 10" → "name Gamer"
 - ◆ "vlan 20" → "name Coder"
3. Assign access ports to VLANs:
 - ◆ "interface range fa0/1 - 2", "switchport mode access", "switchport access vlan 20"
 - ◆ "interface range fa0/3 - 4", "switchport mode access", "switchport access vlan 10"
4. Configure trunk links on inter-switch ports:
 - ◆ "interface fa0/5", "switchport mode trunk"
 - ◆ "interface fa0/6", "switchport mode trunk"
5. On Switch 4, assign VLAN 10 to fa0/1 – fa0/4 and configure trunk to Switch 3:
 - ◆ "interface fa0/5", "switchport mode trunk"
6. Verify VLANs and trunk links using "show vlan brief" and "show interfaces trunk".
7. Save configuration using "write memory".
8. Test connectivity using "ping <destination IP>".

Figures:

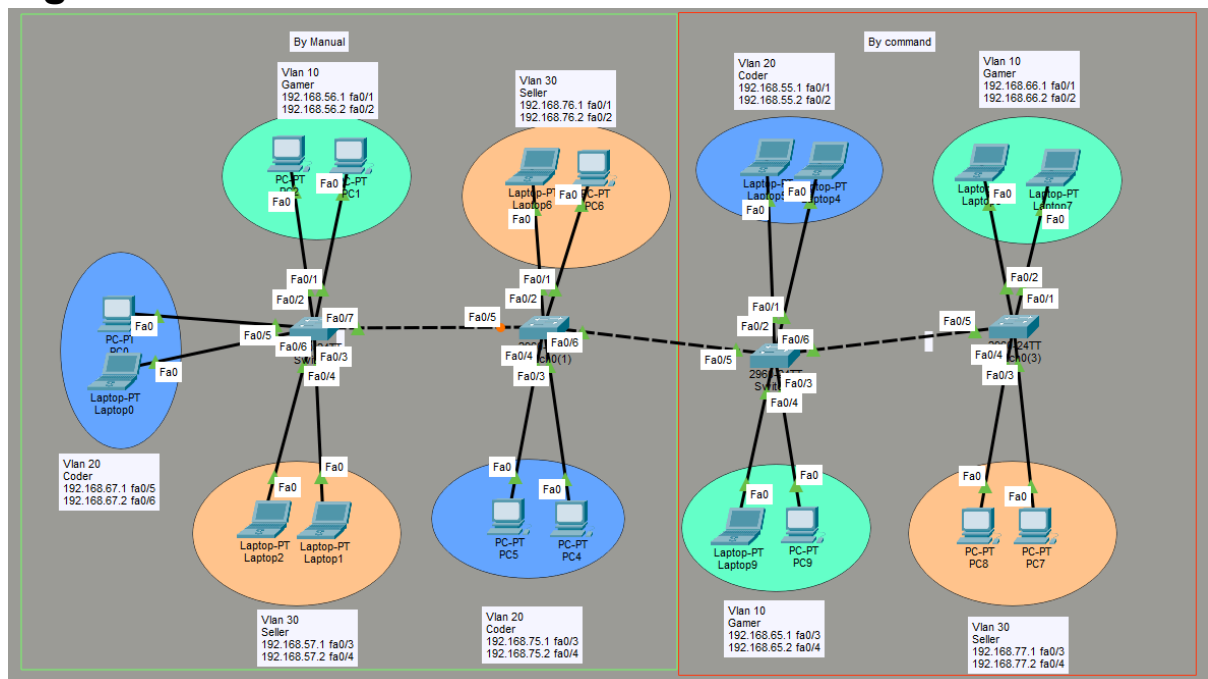


Figure 1: Network topology showing four switches, VLANs, and connected PCs/Laptops.

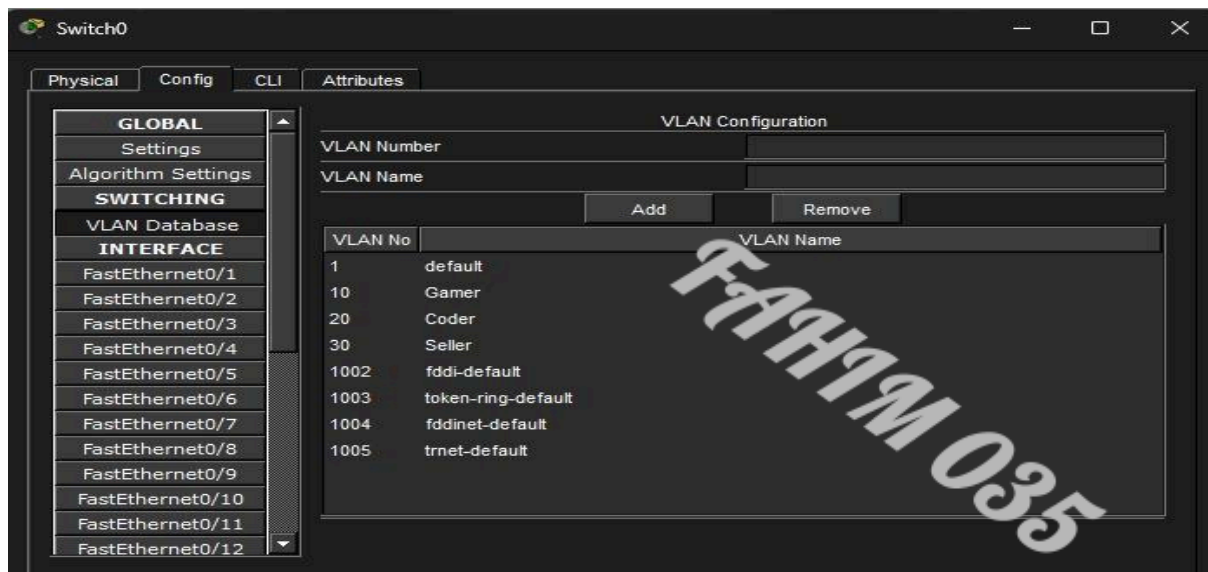


Figure 2: VLAN configuration screenshot for Switch 1 & 2 (Vlan Manual).

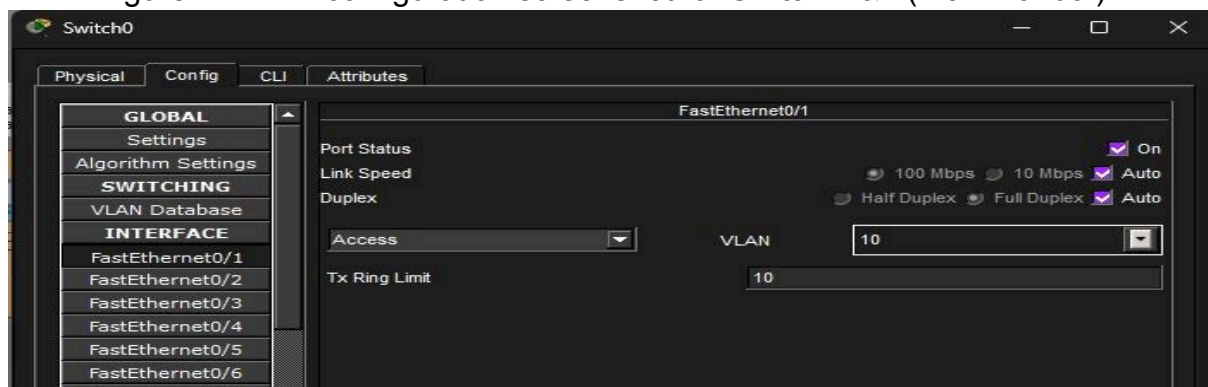


Figure 3: VLAN configuration screenshot for Switch 1 & 2(interface Manual).

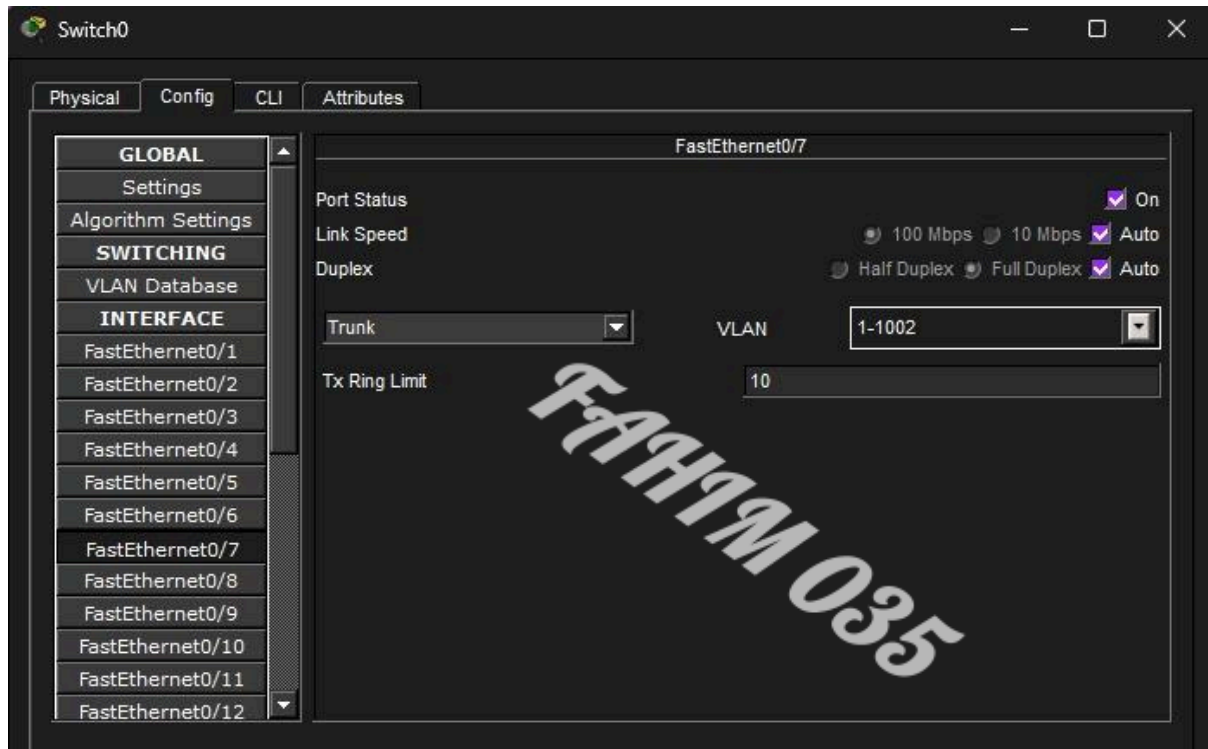


Figure 4: VLAN configuration screenshot for Switch 1 & 2(interface Manual).

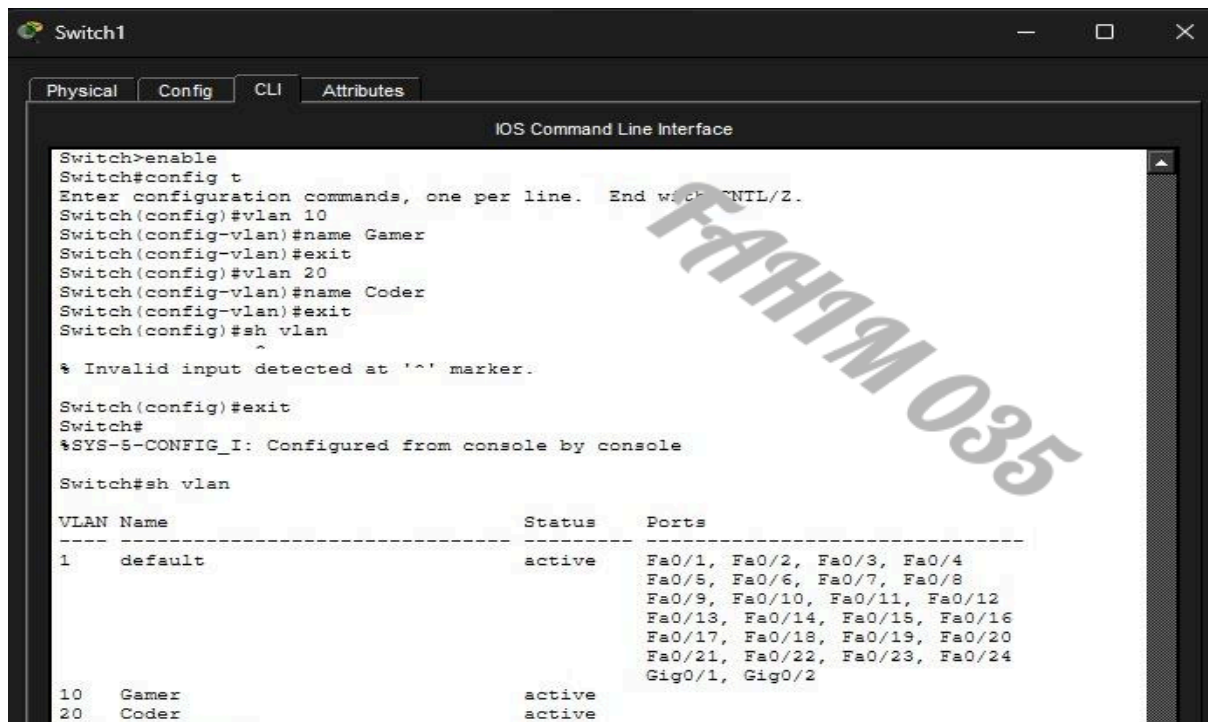


Figure 5: VLAN configuration screenshot for Switch 3 & 4(Command).

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Switch1
Physical Config CLI Attributes
IOS Command Line Interface
Switch(config)#
Switch(config)#
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fa0/5
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
\
^
% Invalid input detected at '^' marker.

Switch(config-if)#exit
Switch(config)#interface fa0/6
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up

```

Figure 6: VLAN configuration screenshot for Switch 3 & 4(Command).

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
	Successful	PC0	Laptop0	ICMP		0.000	N	0	(e...
	Successful	PC2	PC1	ICMP		0.000	N	1	(e...
	Successful	Laptop2	Laptop1	ICMP		0.000	N	2	(e...

Figure 6: VLAN Message passing.

Conclusion:

VLANs were successfully configured across four switches. Devices were logically separated into Gamer, Coder, and Seller VLANs, and trunk links allowed inter-switch communication while maintaining VLAN isolation. The experiment demonstrated VLAN implementation, logical segmentation, and improved network organization and security.